

Yujie Lin

PERSONAL INFORMATION

Emails: lyj23@mails.tsinghua.edu.cn

Languages: Chinese, English

RESEARCH INTERESTS

Low-dimensional topology and symplectic topology, with a particular focus on smooth 4-manifolds, gauge theory, diffeomorphism and symplectomorphism groups, and symplectic dynamics.

EDUCATION

09/2023 - present Ph.D. in Mathematics (expected June 2028) at **Tsinghua University**
 Advisor: Jianfeng Lin
09/2019 - 06/2023 B.S. in Mathematics at **Tsinghua University**
 GPA: 3.72/4.00
09/2016 - 06/2019 High School Diploma, at **Jieyang No.1 Middle School**, Guangdong, China

RESEARCH PAPERS

Yujie Lin and Yi Sha (2026). “On the Homotopy Coherent Nielsen Realization Problem for K3-Type 4-Manifolds”. In: *arXiv preprint arXiv:2606.24482*.

Yujie Lin (2025). “A note on the boundary Dehn twist of $K3$ surfaces”. In: *arXiv preprint arXiv:2506.10444*.

Yunan Wang, Chuxiong Hu, Zeyang Li, Yujie Lin, Shize Lin, and Suqin He (2025). “Chattering Phenomena in Time-Optimal Control for High-Order Chain-of-Integrator Systems With Full State Constraints”. In: *IEEE Transactions on Automatic Control*.

Yunan Wang, Chuxiong Hu, Yujie Lin, Zeyang Li, Shize Lin, and Suqin He (2025). “A Novel State-Centric Necessary Condition for Time-Optimal Control of Controllable Linear Systems Based on Augmented Switching Laws”. In: *IEEE Transactions on Automatic Control*.

INVITED TALKS

- G2T2 Seminar, Yau Mathematical Sciences Center, Tsinghua University, October 2025
- Greater Bay Area Doctoral Student Forum in Mathematics, CUHK, February 2026